Drop-in text for Sec. VI-E (deviation bound) and Sec. VI-B (ramp tracking)

numbering continues from Eq. (86); symbols follow the manuscript

VI-E continuation: impulse response and the bound on e_φ

After Proposition 1 / Eq. (86).

The homogeneous part of (86) is spanned by $\cosh C_2\tau$ and $\sinh C_2\tau$ with $C_2 = \sqrt{Wz_{CoM}/J_P}$ from (20). Its impulse response — the solution of $J_P\ddot{h} - Wz_{CoM}\, h = \delta(\tau)$ from rest — follows from the momentum jump $J_P[\dot{h}(0^+) - \dot{h}(0^-)] = 1$, the position being unable to jump:

$$h_\varphi(\tau) = \frac{\sinh(C_2\tau)}{J_P C_2}, \qquad h_\varphi(0) = 0, \qquad \dot{h}_\varphi(0) = \frac{1}{J_P}. \tag{87}$$

Because $h_\varphi(0) = 0$, the boundary term generated by differentiating a convolution with a variable upper limit,

$$\frac{d}{d\tau}\int_0^\tau h_\varphi(\tau - s)\rho(s)\,ds = \underbrace{h_\varphi(0)}_{=0}\rho(\tau) + \int_0^\tau \dot{h}_\varphi(\tau - s)\rho(s)\,ds,$$

vanishes for *any* forcing. The zero initial conditions in (86) therefore set both homogeneous coefficients to zero — no assumption on $\rho(0)$ is required — and the deviation is the Duhamel integral alone,

$$e_\varphi(\tau) = \int_0^\tau h_\varphi(\tau - s)\,\rho(s)\,ds, \qquad e_\omega(\tau) = \int_0^\tau \dot{h}_\varphi(\tau - s)\,\rho(s)\,ds. \tag{88}$$

The two channels. Two terms of (79) leave the estimator span: the gravity remainder (83) and the bilinear ground-effect coupling (79)(iv),

$$\rho(\tau) = \rho_\varphi(\tau) + \rho_{GE}(\tau), \qquad \rho_{GE}(\tau) = \beta_M \dot{M}\tau\varphi. \tag{89}$$

Both vanish at the onset — with $\varphi_{\text{nom}} \sim \tau^3$ the gravity remainder is $\mathcal{O}(\varphi^2) \sim \tau^6$ and the bilinear term is $\tau \cdot \tau^3 = \tau^4$ — so each is largest at the window end.

An envelope bound that can be verified by hand. The estimator consumes the *rate*: the two-segment fit is applied to $\omega = C_1(\cosh C_2\tau - 1)$ reconstructed from the gyro, and the onset is the argument that minimises its residual. The quantity that must be bounded is therefore e_ω, and e_φ enters only through the gravity remainder that generates ρ. Replacing ρ by a constant $\bar{\rho}$ and integrating the two kernels, $\int_0^\tau \dot{h}_\varphi = \sinh(C_2\tau)/(J_P C_2)$ and $\int_0^\tau h_\varphi = (\cosh C_2\tau - 1)/(J_P C_2^2)$, with $J_P C_2 = Wz_{CoM}/C_2$ and $J_P C_2^2 = Wz_{CoM}$,

$$\big|e_\omega(\tau_{\text{end}})\big| \le \frac{\bar{\rho}\,C_2}{Wz_{CoM}}\sinh x, \qquad \big|e_\varphi(\tau_{\text{end}})\big| \le \frac{\bar{\rho}}{Wz_{CoM}}\big(\cosh x - 1\big), \qquad x \triangleq C_2\tau_{\text{end}}. \tag{90}$$

The inertia cancels in both. Only Wz_{CoM}, C_2, the dimensionless group x and the forcing level enter.

The window length is bounded a priori by the tilt cap. Neither channel maximum nor the kernel factor requires x to be solved for. With the nominal excursion of (20),

$$\varphi_{\text{end}} = \frac{\dot{M}}{Wz_{CoM}C_2}\,\Lambda(x), \qquad \Lambda(x) \triangleq \sinh x - x = \sum_{k\ge1}\frac{x^{2k+1}}{(2k+1)!}. \tag{91}$$

Every term of Λ is positive, so $\Lambda(x) \ge x^3/6$, and since Λ increases, imposing the tilt cap $\varphi_{\text{end}} \le \varphi_{\max}$ inverts in closed form:

$$x \le \bar{x} \triangleq \Big(\frac{6\,\varphi_{\max}Wz_{CoM}C_2}{\dot{M}}\Big)^{1/3}, \qquad \tau_{\text{end}} \le \frac{\bar{x}}{C_2} = \Big(\frac{6\,\varphi_{\max}Wz_{CoM}}{C_2^2\dot{M}}\Big)^{1/3}, \tag{92}$$

where $J_P = Wz_{CoM}/C_2^2$ has been used to remove the inertia. The retained term is the inertia-only rise $\varphi = \dot{M}\tau^3/(6J_P)$ obtained by dropping the destabilising $Wz_{CoM}\varphi$: it is the slowest admissible ascent and therefore reaches the cap latest, which is why (92) bounds x from above and why the worst case is the *slowest* ramp, not the fastest. The moment increment over the window and the coupling channel follow with no measured quantity at all,

$$\Delta M_{\rm win} = \dot{M}\tau_{\rm end} \le \Big(\frac{6\,\varphi_{\max} Wz_{CoM}\dot{M}^2}{C_2^2}\Big)^{1/3}, \qquad \rho_{GE,\max} = |\beta_M|\,\Delta M_{\rm win}\varphi_{\rm end} \le |\beta_M|\,\varphi_{\max}\Big(\frac{6\,\varphi_{\max} Wz_{CoM}\dot{M}^2}{C_2^2}\Big)^{1/3}, \tag{93}$$

and the kernel factors of (90) close on the same constant through the identity $\cosh x - 1 = \Lambda(x) + (x - 1 + e^{-x})$ with $0 \le x - 1 + e^{-x} < x$, together with $\sinh x = \Lambda(x) + x$:

$$\cosh x - 1 \;\le\; \sinh x \;\le\; \Lambda(x) + \bar{x} \;\le\; \frac{\bar{x}^3}{6} + \bar{x}. \tag{94}$$

Both kernels therefore share one bracket.[1]

Why the envelope alone does not settle the question. Nothing beyond $|\rho| \le \rho_{\max}$ enters (90): no monotonicity, no shape, no property of the trajectory but the window length. That generality is also its cost. A constant envelope spreads the forcing over the whole window, whereas both channels are concentrated at its end (τ^6 and τ^4), and the mismatch grows with x, i.e. with the *slowest* ramps. Evaluated with $\bar{\rho} = \rho_{\max}$ over the identified constants, (90) returns a relative deviation of 446% at $\dot{M} = 0.10$ N m/s — larger than the signal it bounds. The envelope form is a sanity check on the algebra, not a result.

What restores it is the one structural fact available for free: ρ rises monotonically from zero at the onset while the kernel $\cosh\big(C_2(\tau - s)\big)$ decreases in s. For functions of *opposite* monotonicity Chebyshev's integral inequality runs in the direction needed, and each maximum in (90) may be replaced by the channel's time average. With the nominal shapes $\rho_\varphi \propto \Lambda(u)^2$ and $\rho_{GE} \propto u\,\Lambda(u)$, $u = C_2\tau$,

$$\bar{\rho} = R_\varphi(x)\,\rho_{\varphi,\max} + R_{GE}(x)\,\rho_{GE,\max}, \qquad R_\varphi \le \tfrac{1}{7}, \qquad R_{GE} \le \tfrac{1}{5}, \tag{95}$$

the limits being the $x \to 0$ values where the two channels are sixth and fourth order; over the flown $x \in [1.79, 5.20]$ they are 0.089–0.133 and 0.145–0.191. The same evaluation that gave 446% returns 41.8% with (95) in place.

Relative form: the result, free of x. Dividing the rate bound of (90) by the nominal rate the estimator fits, $\omega_{\rm nom,end} = C_1(\cosh x - 1)$ with $C_1 = \dot{M}/Wz_{CoM}$, and writing $\Delta M_{\rm win} = \dot{M}\tau_{\rm end} = \dot{M}x/C_2$ for the moment increment over the window,

$$\frac{|e_\omega|_{\rm end}}{\omega_{\rm nom,end}} \le \frac{\Psi(x)\,\bar{\rho}}{\Delta M_{\rm win}}, \qquad \Psi(x) = \frac{x\sinh x}{\cosh x - 1} = x\coth\frac{x}{2}, \tag{96}$$

the closed form following from $\cosh x - 1 = 2\sinh^2(x/2)$ and $\sinh x = 2\sinh(x/2)\cosh(x/2)$; $\Psi \ge 2$ everywhere, which is the inequality the exponent-perturbation argument of Sec. VI-D already uses. Here the two approximations work against each other: Ψ grows with x while the averages of (95) shrink with it, and each product is bounded uniformly — ΨR_φ rises monotonically from 2/7 to 1/2 and ΨR_{GE} from 2/5 to 1. Hence

$$\boxed{\frac{|e_\omega|_{\rm end}}{\omega_{\rm nom,end}} \;\le\; \frac{\frac{1}{2}\rho_{\varphi,\max} + \rho_{GE,\max}}{\Delta M_{\rm win}}, \qquad |\Delta M_{\rm crit}| \;\le\; \frac{\rho_{\varphi,\max}}{7} + \frac{\rho_{GE,\max}}{5}} \tag{97}$$

independently of x, C_2, J_P and Wz_{CoM}: the window length that (92) went to some trouble to bound does not survive into the conclusion, and is needed only to bound $\rho_{GE,\max}$ through (93). The second inequality follows because the onset minimises the two-segment residual: linearising its stationarity condition gives $\Delta M_{\rm crit} = -\int_0^{\tau_{\rm end}} \rho\, w$ with a non-negative weight normalised to $\int w = 1$, so $|\Delta M_{\rm crit}| \le \bar{\rho}$. A reviewer can evaluate (97) from numbers already in the paper.

[1] The step matters. $\Lambda(x)$ is known *as an equality* — the tilt cap supplies it — so only the residual linear term needs $\bar{x}$. Substituting $\bar{x}$ into sinh instead is exponentially lossy: at the slowest ramp $\bar{x} = 7.23$ gives $\sinh\bar{x} = 691$ against $\Lambda + \bar{x} = 70.3$, a factor of nine thrown away for nothing.

No pivot inertia is assumed anywhere. C_2 is not computed from J_P; the two-stage calibration of Sec. V identifies the pair (C_2, K) directly from the onset fits, and $Wz_{CoM} = 1/K$ is one of them. $J_P = 1/(KC_2^2)$ is a derived by-product that enters none of (90)–(97) — which is as well, since the identified J_P spans 0.10–0.28 kg m^2 against the rigid parallel-axis value $J_{CoM} + md^2 \approx 0.43$ kg m^2, a gap the compliant contact makes uninterpretable as an inertia. Over the ten configurations the identified constants span $C_2 \in [3.52, 5.67]$ s^{-1} and $Wz_{CoM} \in [2.96, 6.65]$ N m, so with $\varphi_{\max} = 10°$ and $\dot{M} \in [0.10, 1.20]$ N m/s, (92) gives $\bar{x} \le 7.23$ at the slowest ramp and (93) gives $\rho_{GE,\max} \le 4.5$ mN m at the fastest — the two extremes lie at opposite corners of the box — against $\rho_{\varphi,\max} \le \frac{1}{2}W(l_p + \lambda_{\text{off}})\varphi_{\max}^2 = 77.0$ mN m (roll; 62.5 pitch), or 50.7 mN m if the stabilising $-Wz_{CoM}\sin\varphi$ term of G'' is retained at the cap. Because $\bar{x}$ depends on the constants only as $(Wz_{CoM}C_2/\dot{M})^{1/3}$, a factor-of-two error in Wz_{CoM} moves it by 26%: the chain is least sensitive to exactly the constants that are least certain.

Evaluation. Every quantity must be read off the *same* window: $\rho_{\varphi,\max}$ and $\rho_{GE,\max}$ at the excursion that window reaches, ΔM_{win} as its own moment increment. Over the 140 gate-passing runs, with $W = 31.59$ N, $l_p + \lambda_{\text{off}} = 0.160/0.130$ m (roll/pitch), $z_{CoM} = 0.30$ m and $\beta_M = 0.0345$ rad^{-1}:

$\dot{M}$ [N m/s]	0.10	0.30	0.65	1.20	median	worst
$\lvert\varphi\rvert_{\text{end}}$ [deg]	5.23	5.90	6.55	6.17	5.95	9.33
ΔM_{win} [N m]	0.072	0.165	0.306	0.479	0.221	—
$\rho_{\varphi,\max}$ [mN m]	19.7	22.5	30.4	27.2	23.6	58.3
$\rho_{GE,\max}$ [mN m]	0.24	0.59	1.24	1.85	0.68	3.00
$x = C_2\tau_{\text{end}}$	3.47	2.64	2.15	1.79	2.67	5.20
$\lvert e_\omega\rvert/\omega_{\text{nom}}$ [%]	13.8	6.0	4.7	2.9	4.9	25.1
$\lvert\Delta M_{\text{crit}}\rvert$ [mN m]	1.94	2.73	3.77	3.76	3.01	7.86
$\lvert\Delta\lambda_{\text{off}}\rvert$ [mm]	0.061	0.086	0.119	0.119	0.095	0.249
ground-effect share of $\bar{\rho}$ [%]	1.8	4.0	6.0	8.8	5.1	25.6

These are bound evaluations, not measurements: each row forwards the *unprojected* channel maximum through (96). Using instead the x-free constants of (97) the same runs give $|\Delta M_{\text{crit}}| \le 8.8$ mN m, i.e. $|\Delta\lambda_{\text{off}}| \le 0.28$ mm — the price of not evaluating R_φ, R_{GE} is under 12%. Applied a priori at the tilt cap rather than at the flown excursions, (97) gives 11.9 mN m and 0.38 mm, which is the guarantee the method carries into a configuration it has not seen.

The per-run propagation reported with the error budget is a separate and much smaller number, because it convolves the *measured* ρ with the exact kernel after the estimator has absorbed its own span. Over the same runs the residual forcing that actually reaches the onset is 2.75 mN m in the median (12.28 at worst) on the gravity channel and 0.53 (3.20) on the ground-effect channel, the rate deviation $|e_\omega|/\omega_{\text{nom}}$ is 0.24% in the median (6.5% at worst summed over channels), and the resulting threshold error is 0.04 mN m in the median and 0.15 at worst — 0.001 to 0.004 mm of offset. The two differ by roughly a factor of 50, which is the price the envelope form pays for assuming nothing about ρ but its size.

The bilinear ground-effect channel contributes 5.1% of $\bar{\rho}$ in the median and 25.6% at worst; it grows with the ramp rate because $\rho_{GE} \propto \dot{M}\tau$, whereas the gravity remainder depends on the excursion alone. Its own contribution to the threshold is 0.00 mN m in the median and 0.01 at worst once propagated, so the channel the reviewer asks about is not the one that sets the budget. The threshold bound $|\Delta\lambda_{\text{off}}| \le 0.25$ mm is the figure that matters: it does not involve ΔM_{win}, and so does not inherit the loss the relative normalisation suffers at slow ramps. The relative bound is loosest at 0.10 N m/s, where the window closes on a small moment increment while the excursion is unchanged — the same ordering the excitation design reports — and it is conservative by roughly a factor 5 overall, since it forwards the *unprojected* remainder (23.6–58.3 mN m here, against 1.2–12.3 mN m measured after the estimator absorbs its own degrees of freedom).

VI-B: ramp input tracking (rate limits)

The rotor response identified in Sec. V (Table 4: $\omega_{n,\text{rot}} = 21.74$ rad/s, $\zeta = 0.7814$, $\tau_d = 15.67$ ms) tracks a ramp with the steady-state lag

$$t_{\text{lag}} = \frac{2\zeta}{\omega_{n,\text{rot}}} + \tau_d = 71.9 + 15.7 = 87.6 \text{ ms},$$

so the realised moment trails the command by $\dot{M}\, t_{\text{lag}}$, i.e. 8.8 mN m at 0.10 N m/s and 105 mN m at 1.20 N m/s.

This lag does *not* bias M_{crit}. The identification consumes the *realised* moment, reconstructed from the measured rotor speeds, not the commanded one; a common lag therefore shifts the moment trace and the rate trace together and leaves the moment read at the detected onset unchanged. What the lag does consume is window length, which the sample-count gate enforces.

Two residual effects remain and are both gated. First, the realised *slope* may differ from the commanded rate; the run-level gate rejects $|\dot{M}_{\text{meas}} - \dot{M}_{\text{cmd}}|/\dot{M}_{\text{cmd}} > 3\%$, and since $C_1 = K\dot{M}$, a 3% rate error perturbs the amplitude constant by 3% — well inside the ±20% mis-specification the sensitivity analysis shows to be harmless. Second, the start of the ramp is not linear while the rotor transient decays; with $4/(\zeta\omega_{n,\text{rot}}) =$ 235 ms to 2%, that transient is confined to the low-moment part of the window, which the slope gate excludes by scoring only the segment above the per-axis moment floor. The linearity gate (30 mN m RMSE about the best-fit line) then acts as a fault detector for ramps that are stepped or aborted outright.